Fast Chroma Prediction Mode Decision based on Luma Prediction Mode for AV1 Intra Coding

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***Abstract*—AV1 is an emerging open-source and royalty-free video compression format, which is jointly developed and finalized in June 2018 by the Alliance for Open Media (AOMedia) industry consortium. AV1 achieves significant improvements in coding efficiency compared with HECV and VP9 thanks to new coding tools. However, encoding complexity is extremely high by selecting best mode among new features. This paper proposes a fast chroma prediction mode decision algorithm based on luma prediction mode for AV1 intra coding. Experiment results show that the proposed algorithm achieves an 15.86% TS with 0.44% BDBR increase compared with AV1 reference software.**

Keywords*—*AV1, fast chroma intra mode decision

I. Introduction

AV1 is an open-source and royalty-free video coding format, which was developed by the AOMedia industry consortium and released in June 2018 as the state-of-the-art in video coding [1]. AV1 employs flexible block partitioning from 128x128 to 4x4, new intra predictions, constrained directional enhancement filter (CDEF) and so on [2]. Through these efforts, AV1 can achieve significant bit rate reduction over the state art of the codecs such as VP9 and HEVC [3][4].

AOM has released reference software for AV1 encoder and decoder [5]. The software has made a lot of efforts to speed up the encoding such as fast encoding algorithms, single instruction multiple data (SIMD) and multi-threading. Despite these efforts, the encoding speed is still slow. This encoder complexity comes from rate distortion optimization (RDO) to find the optimal mode and partition size among numerous coding modes. A high-speed encoding is desirable for practical application. This paper focuses on fast intra encoding because AV1 employs a lot of new coding tools for intra coding.

To exploit more varieties of spatial redundancy in directional textures, AV1 adopts 56 directional modes, extended smooth modes, Paeth predictor, intra block copy, palette mode and chroma predicted from luma (CfL). Unlike conventional codecs such as H.264/AVC and HEVC, most intra prediction (IP) modes used in luma signals are also applied to chroma signals [6][7]. Therefore finding the optimal IP mode of chroma requires a large amount of computation due to the variety of chroma IP modes. However, since the video standards prior to AV1 has fewer IP modes for chroma signal, most existing algorithms have focused on speeding up IP mode decision of luma signals. Therefore, a new algorithm for determining the IP mode of the chroma signal at high speed for AV1 is needed.

In this paper, we use the correlation between IP modes of luma and chroma signals to decide the IP mode of chroma signal. Luma and chroma signals are known to have some degree of correlation and there are various compression algorithms using them [8-10]. In video codec standard, HEVC has a direct mode that allows you to use the same mode as luma mode in chroma mode [7]. CfL prediction in AV1 is a chroma-only intra predictor that models chroma pixels as a linear function of the coincident reconstructed luma pixels. Based on this relationship, this paper proposes a method to reduce the candidates of IP mode of chroma.

The remainder of this paper is organized as follows. Section 2 shows the relationship between IP modes of luma and chroma signals through experiments. Section 3 introduces the proposed fast IP mode decision of chroma based on the relationship. In section 4, experiments are carried out to demonstrate the efficiency of the proposed algorithms. Section 5 concludes this paper.

II. The Relationship between Luma and Chroma Intra Prediction Modes in AV1

Table I shows IP modes for the luma and chroma signals in AV1. This table shows only main directional modes from DC to D67, and each main direction has additional 6 delta angles. The INTRA FILTER in Table I contains five intra-filter modes: DC, H, V, D157, and PAETH. This is shown in Fig. 1 (b) as FILTER_DC, FILTER_H, FILTER_V, FILTER_D157, and FILTER_PAETH. In the chroma signal, Cb and Cr share the same IP mode. Therefore, one intra block requires one luma IP mode and one chroma IP mode. In Table I, the difference between IP modes of luma and chroma is INTRA_FILTER and CfL mode. INTRA_FILTER modes are only applied to the luma signal, and the CfL mode is only applied to the chroma signal. This similarity of the modes allows luma IP mode to help determine the chroma IP mode directly.

In order to investigate the relationship between luma and chroma IP modes, the distribution of the two modes is obtained through experiments. The experiments are conducted with 100 frames from 5 sequences with FHD resolution. And all frames are encoded intra frame with QP42. Fig. 1 shows the distribution of the chroma IP mode for each luma IP mode averaged over the results of five sequences. The x-axis represents the chroma IP mode and the y-axis represents the probability. For example, in Fig. 1(a), if the luma IP mode is selected as H, the probability at which H is selected in chroma IP mode is 0.5.

Fig. 1 (a) and (b) show the probability distribution in the directional mode and the non-directional mode of luma IP,

TABLE I. INTRA PREDICTION MODES FOR LUMA AND CHROMA SIGNALS

Prediction modes	*Luma signal*	*Chroma signal*
DC	Yes	Yes
V	Yes	Yes
H	Yes	Yes
D45	Yes	Yes
D135	Yes	Yes
D113	Yes	Yes
D157	Yes	Yes
D203	Yes	Yes
D67	Yes	Yes
SMOOTH	Yes	Yes
SMOOTH-H	Yes	Yes
SMOOTH-V	Yes	Yes
PAETH	Yes	Yes
INTRA FILTER	Yes	No
CfL	No	Yes
INTRA BLOCK COPY	Yes	Yes
PALLETE	Yes	Yes

respectively. From these figures we can observe three phenomena:

(a) It is very unlikely that the direction of the luma IP mode and the direction of chroma IP mode are different.

(b) Regardless of the luma IP mode, there is a high probability that DC, SMOOTH, or CfL are selected for the chroma signal.

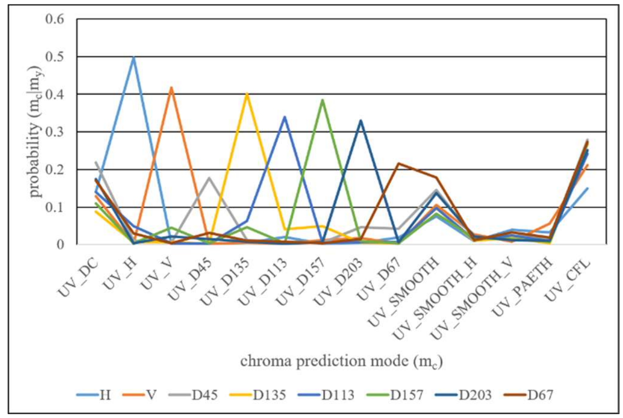
(c) If the luma IP mode is PAETH or INTRA_FILTER mode, it is highly likely that the PAETH mode is selected as the IP mode of the chroma signal.

From above four observations, we propose the following chroma IP mode decision algorithm.

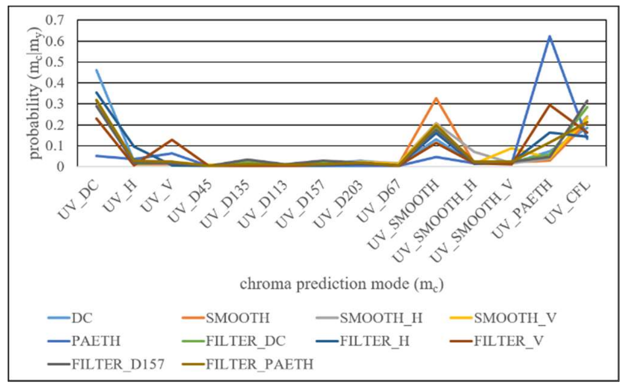
III. PROPOSED ALGORITHM

After determining the optimal IP mode (Y_MODE) for the luma signal in the current block, the proposed method determines the IP mode for the chroma signal based on Y_MODE. There are a total of 15 candidate IP modes for chroma signal. Rate distortion optimization (RDO) is performed for each candidate mode, and the mode with the smallest rate distortion (RD) cost is selected as the best IP mode. We reduce candidate modes by eliminating unnecessary candidate modes based on Y_MODE to achieve fast chroma IP mode decision.

The selection of candidate modes is derived from the observation of Section II. From the observation, regardless of the type of Y_Mode, DC, SMOOTH, and CfL are selected with high probability in the chroma signal. Therefore, DC, SMOOTH, and CfL are always included in candidates by default. If the Y_Mode is a directional mode, the probability that a chroma IP mode having a different direction from Y_MODE is selected is very low. Therefore, only chroma IP mode with the same direction as Y_MODE is added to candidate modes. The optimal delta angle for the main direction for chroma is recalculated. From the observation, if Y_MODE is one of the INTRA_FILTER or PAETH mode, the probability that the chroma mode will be selected as PAETH mode is high. Therefore, if Y_MODE is one of the INTRA_FILTER or PAETH mode, PAETH mode is added to the candidate modes.



(a)



(b)

Fig.1. Distribution of chroma IP mode according to IP mode of luma signal. (a) In case of the luma IP mode is directional mode (b) In case of the luma IP mode is non-directional mode

To further reduce the number of candidate modes, we selectively remove CfL in candidate modes. CfL is a chroma-only intra predictor that models chroma pixels as a linear function of coincident reconstructed luma pixels. CfL determines the parameters of the linear function based on the original chroma pixels and signals them in the bitstream. To encode CfL, 3 parameters such as ***cfl alpha_signs***, ***cfl alpha_u***, and ***cfl_alpha_v*** have to be transmitted. Therefore, coding a simple area with CfL is inefficient because it requires additional 3 parameters to be encoded. Therefore, we apply CfL only to the complex region. Whether or not the current block is a complex area is determined by the block variance of luma. If luma's block variance is smaller than a certain threshold, CfL is removed to the candidate modes.

Fig.2 shows the procedure for determining IP candidate modes of chroma signal. First, the initial candidate mode includes only DC and SMOOTH. If luma block's variance is greater than a pre-defined threshold, CfL is added to candidate modes. If Y_MODE is one of the directional mode, the directional mode of Y_MODE is added to the candidate modes. Or if Y_MODE is one of INTRA_FILTER or PAETH mode, PAETH mode is added to candidate modes. As a result, the number of final candidate modes can range from a minimum of 2 to 4, depending on the characteristics of the luma signal. Therefore, the computational complexity can be significantly reduced for chroma IP mode decision by performing RDO only for the few candidate modes.

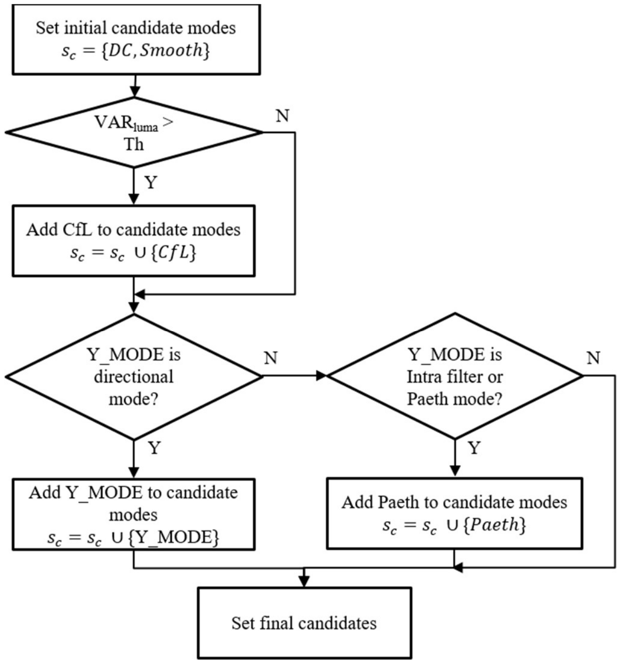


Fig.2. The procedure for determining IP candidate modes of chroma signal

IV. Experimental Results

To evaluate the performance of the proposed algorithms, it was implemented in AV1 reference software [5]. The 4K , 2K, and 720P test sequences are tested for the simulation [11][12]. Since we focus on the performance of intra coding, experiments are carried out with the all-intra-frame. To reduce the encoding time, we set the speed feature preset of encoder to 4. QP values of 27, 36, 45 and 54 are used for the evaluation. For coding efficiency is measured with Bjontegaard difference bitrate (BDBR) [13] and computational complexity is measured with encoding time saving (TS) as (1). The reference software runs on two CPUs @ 3.4GHz (32 cores) with 64 GB memory.

$$\mathrm{TS}\ (\%) = \frac{T_{aom} - T_{proposed}}{T_{aom}} \times 100 \qquad (1)$$

Table II shows the performance of the proposed algorithm. On average, the proposed algorithm achieves an 15.86% TS with 0.44% BDBR increase. The minimum TS of the proposed algorithm is 14.09% of Parkscene sequence, and the performance variation of TS is small. This shows that the proposed algorithm can achieve uniform speed improvement regardless of image characteristics and resolution.

V. Conclusion

In this paper, we proposed a fast chroma IP mode decision algorithm for AV1 intra coding. This paper reduced the candidate modes of chrome IP mode decision by using luma 's IP mode and block variance. The computational complexity is significantly reduced by performing RDO only for the few candidate modes. Experiment results show that the proposed algorithm achieves an 15.86% TS with 0.44% BDBR increase compared with reference software. The proposed algorithm is expected to be applied to another codecs with little difference between luma IP mode and chroma IP mode. We plan to apply the proposed algorithm to the Inter frame of AV1 in the near future.

TABLE II. Encoding Results of the Proposed Algorithm

Size	Sequence	BDBR (%)	TS (%)
UHD (3840x2160)	Beauty	0.10	16.96
	Jockey	0.32	16.34
	ShakeNDry	0.07	15.49
	YachtRide	0.53	15.81
FHD (19201080)	Kimono	0.23	16.63
	ParkScene	0.13	14.09
	Cactus	0.41	15.18
	BasketballDrive	0.94	15.93
	BQTerrace	0.32	15.63
HD (1280x720)	Fourpeople	0.58	16.65
	Johnny	1.19	16.83
	Average	0.44	15.86

Acknowledgment

"This work was supported by Institute for Information & communications Technology Promotion (IITP) grant funded by the Korea government (MSIT) (No. 2017-00307, Development of Tiled Streaming Technology for High Quality VR Contents Real-Time Service)."

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